Project Description on

Hotel Reservation System

Course Code :SWE-223
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Hotel Reservation System

Introduction

The Hotel Reservation System is a Java-based application developed to manage hotel activities such as room management, customer management, and reservations. The main purpose of this project is to make hotel record keeping easier and more organized compared to manual methods.

This system allows users to store and manage information about rooms, customers, and bookings in a proper way. It uses Object-Oriented Programming (OOP) concepts, along with file handling and serialization, so that data is saved permanently and not lost when the program is closed. The project also uses a Date class for handling check-in and check-out dates and Enum classes to manage fixed values like room type and reservation status.

Objectives of the Project

The main objectives of the Hotel Reservation System are:

- To make the management of hotel rooms, customers, and reservations automatic and organized.
- To reduce manual work and avoid mistakes in record keeping.
- To provide an easy-to-use system for adding, updating, deleting, and searching data.
- To apply Java Object-Oriented Programming (OOP) concepts in a real-life project.
- To store data permanently using file handling and serialization.
- To design the system in a modular way so it is easy to understand and maintain.

System Overview

The Hotel Reservation System is divided into three main modules to make the system easy to understand and manage.

Room Management

The Room Management Module is used to manage all room-related information. Each room has a unique room ID and includes details such as room name, room type, capacity, price, and room status. The room status shows whether a room is available, reserved, or occupied.

Using this module, users can add new rooms, view room details, update room information, and delete rooms when needed. This module helps keep accurate and up-to-date records of room availability.

Customer Management

The Customer Management Module is used to manage customer information. Each customer record includes customer ID, name, email, phone number, and address.

This module allows users to add new customers, update customer details, search for customers, and delete customer records. Keeping customer information organized helps provide better service and ensures correct reservation processing.

Reservation Management Module

The Reservation Management Module is responsible for managing bookings. It connects customers with rooms and stores details such as reservation ID, selected room, customer information, check-in date, check-out date, number of nights, number of guests, and reservation status.

The reservation status can be Confirmed, Cancelled, or Completed. This module helps track reservations properly and prevents double booking or room conflicts.

System Design

The system follows a modular design, where each entity is represented by a separate class. The main classes used in the project include:

- **Room Class:** Stores room-related information and methods.
- **Customer Class:** Stores customer details.
- **Reservation Class:** This class stores all reservation details and connects a customer with a room.
- **Date Class:** A simple custom Date class is used to store check-in and check-out dates. It keeps the day, month, and year separately, making date handling easy to understand and avoiding confusion.
- **Enum Classes:** Enum classes are used to store fixed values such as room type, room status, and reservation status. Using enums helps prevent wrong values, makes the code easy to read, and keeps the system more organized.
- **Date Class:** Handles date-related data such as check-in and check-out dates.
- **FileHandler Class:** Manages reading and writing serialized objects to files.

This design improves code readability, reusability, and maintainability. Any future changes can be made easily without affecting the entire system.

Data Storage and File Handling

The project uses Java file handling with serialization to store data permanently. Separate files are maintained for rooms, customers, and reservations. When the application starts, existing data is loaded from the files. When changes are made, updated objects are saved back to the files.

This ensures:

- Data persistence
- No data loss after program termination
- Simple and efficient storage mechanism

The FileHandler class handles all file-related operations such as saving, loading, updating, and deleting records.

Graphical User Interface (GUI)

The Hotel Reservation System uses Java Swing to create a simple and user-friendly Graphical User Interface (GUI). Swing components such as buttons, labels, text fields, tables, and combo boxes are used to allow users to interact with the system easily.

The main interface provides options to manage rooms, customers, and reservations through separate screens for adding, viewing, updating, and deleting records. Input fields and dropdown menus make data entry simple and help reduce errors.

Using Swing, the GUI is clear, interactive, and easy to navigate, making the system more efficient and improving the overall user experience.

ER Diagram (Entity-Relationship Diagram)

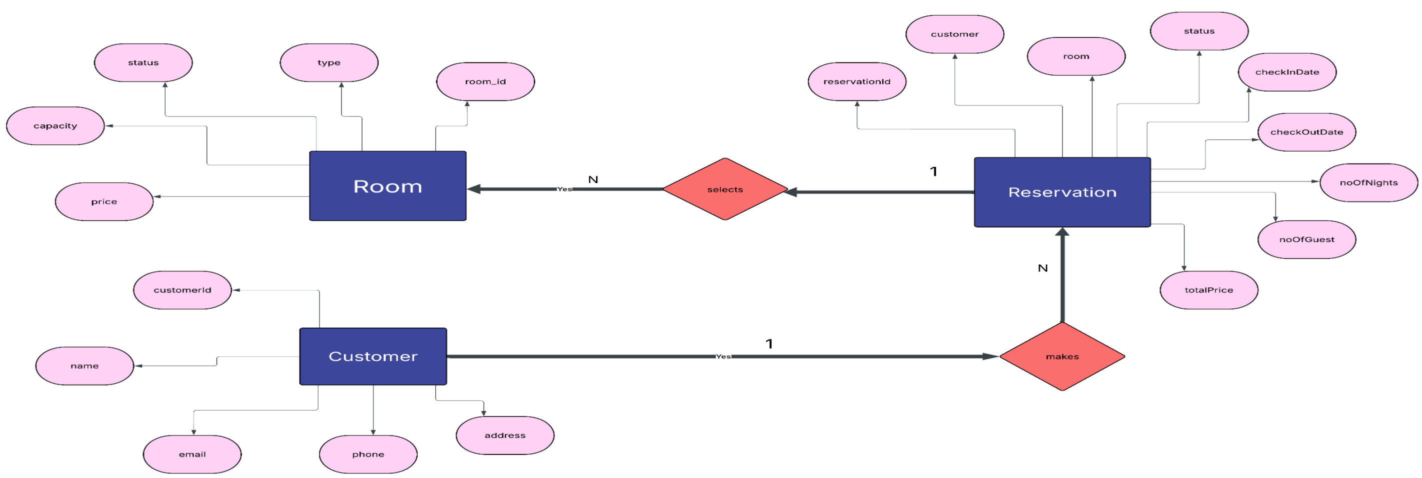
The **ER Diagram** shows how the main entities of the hotel reservation system are related to each other. The main entities are:

1. **Room** – stores information about each hotel room, such as room ID, room name, room type, capacity, price, and status.
2. **Customer** – stores details about customers, including customer ID, name, email, phone number, and address.

3. **Reservation** – connects a customer with a room and stores reservation details like reservation ID, check-in and check-out dates, number of nights, number of guests, and reservation status.

Relationships:

- A **Customer** can have **many Reservations**.
- A **Room** can be part of **many Reservations**, but a room can only be booked for one reservation at a time.



Class Diagram

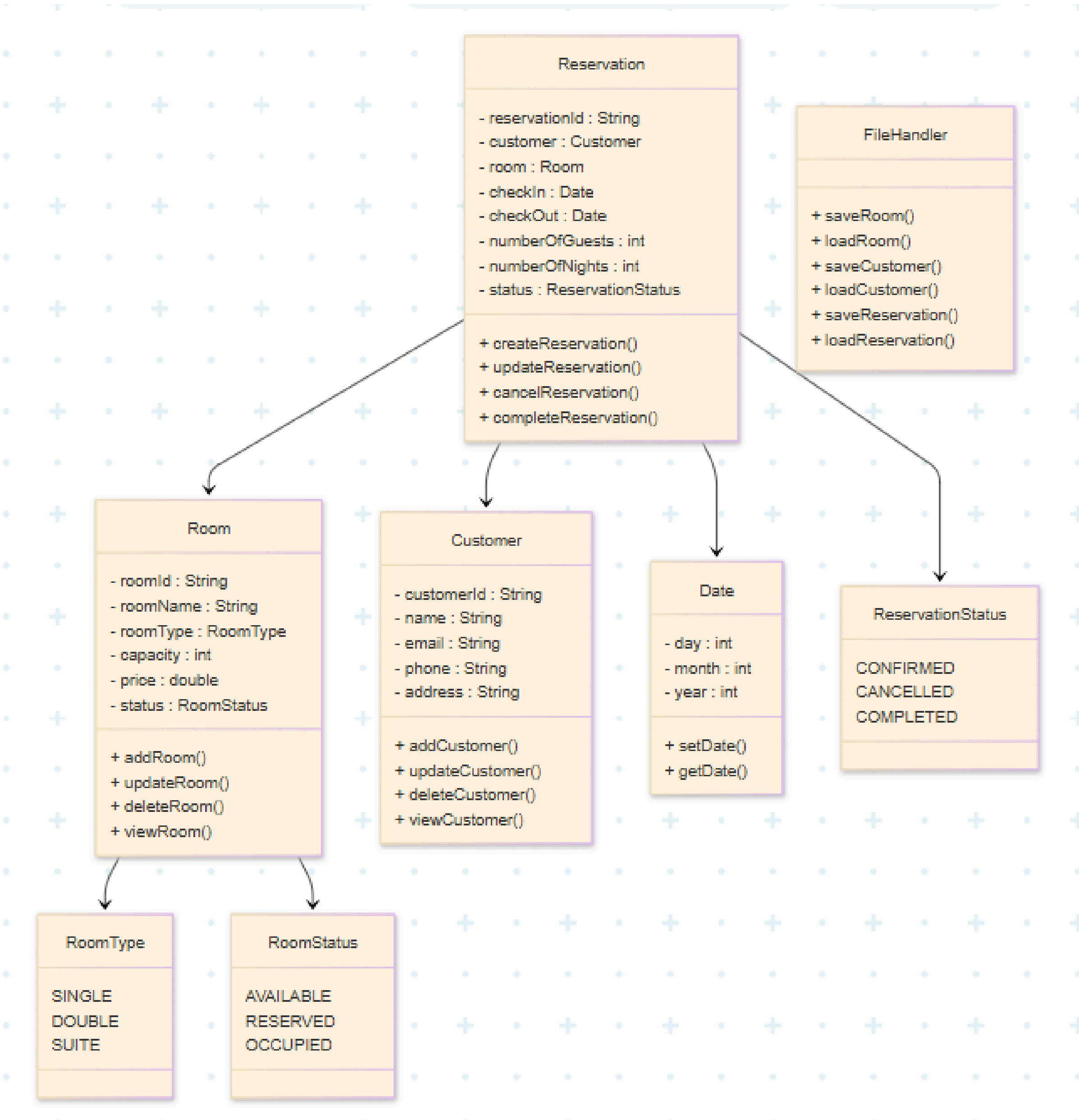
The Class Diagram shows the structure of the system and how different classes are connected. The main classes are:

1. **Room Class** – stores room details and methods to add, update, view, or delete rooms.
2. **Customer Class** – stores customer information and provides methods to manage customers.

3. **Reservation Class** – stores reservation information and links rooms with customers.
4. **Date Class** – stores check-in and check-out dates (day, month, year).
5. **Enum Classes** – define fixed values for room type, room status, and reservation status.
6. **FileHandler Class** – handles saving, reading, updating, and deleting data from files using Java serialization.

Connections:

- Reservation Class uses Room and Customer classes.
- Reservation Class also uses Date Class for reservation dates.
- Enum Classes are used in Room and Reservation to store predefined values.
- FileHandler is used by all classes to manage file operations.


Reservation
- reservationId : String
- customer : Customer
- room : Room
- checkIn : Date
- checkOut : Date
- numberOfGuests : int
- numberOfNights : int
- status : ReservationStatus
+ createReservation()
+ updateReservation()
+ cancelReservation()
+ completeReservation()
FileHandler
+ saveRoom()
+ loadRoom()
+ saveCustomer()
+ loadCustomer()
+ saveReservation()
+ loadReservation()
Room
- roomId : String
- roomName : String
- roomType : RoomType
- capacity : int
- price : double
- status : RoomStatus
+ addRoom()
+ updateRoom()
+ deleteRoom()
+ viewRoom()
Customer
- customerId : String
- name : String
- email : String
- phone : String
- address : String
+ addCustomer()
+ updateCustomer()
+ deleteCustomer()
+ viewCustomer()
Date
- day : int
- month : int
- year : int
+ setDate()
+ getDate()
ReservationStatus
CONFIRMED
CANCELLED
COMPLETED
RoomType
SINGLE
DOUBLE
SUITE
RoomStatus
AVAILABLE
RESERVED
OCCUPIED